

Practical Task-Software Architecture Patterns

The Architecture patterns covered in the learning content:

- Layered Architecture
- Repository Architecture
- Client-Server Architecture
- Pipe and Filter Architecture

Architecture Pattern	Real-world example	When should it be used?	Why is it appropriate?
Layered Architecture	Banking System (Online Banking Application)	When different parts of the program have different jobs.	Each layer has its own job. It is easy to update or fix one part without changing everything
Repository Architecture	GitHub	When many people or programs need to use the same data.	All the data is stored in one place. Everyone works with the same information.
Client-Server Architecture	Gmail	When users need to connect to a server over the internet.	The client sends a request. The server processes the request and sends back the information.
Pipe and Filter Architecture	Photoshop	When data needs to go through several steps.	Each filter does one job. The output from one filter goes to the next filter.

Systems that Combine Two or More Patterns

Example 1:

Netflix

Patterns used	Strengths	Limitations
Client-Server Layered	Works well for many users. Easy to update. Easy to maintain.	Needs a good internet connection. If the server has a problem, users may not be able to watch videos.

Example 2:

Amazon

Patterns used	Strengths	Limitations
Client-Server Repository	Stores all product information in one place. Many users can shop at the same time. Data is easy to manage.	If the database has a problem, the website may not work properly. Needs strong security to protect customer information.